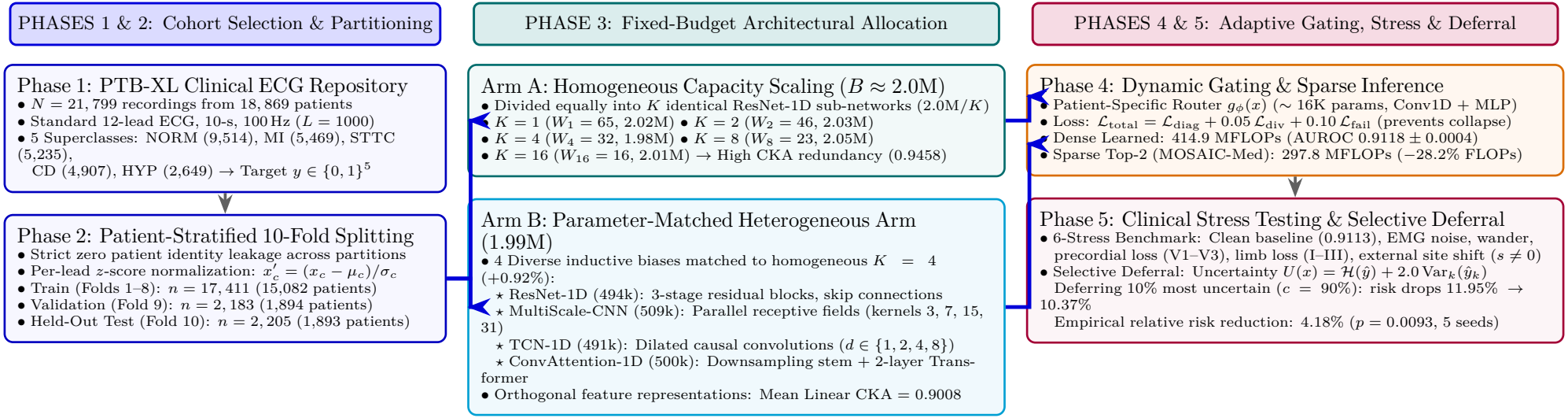
PHASES 1 & 2: Cohort Selection & Partitioning
Phase 1: PTB-XL Clinical ECG Repository
• $N = 21,799$ recordings from 18,869 patients
• Standard 12-lead ECG, 10-s, 100 Hz ($L = 1000$)
• 5 Superclasses: NORM (9,514), MI (5,469), STTC (5,235),
CD (4,907), HYP (2,649) → Target $y \in \{0,1\}^5$
Phase 2: Patient-Stratified 10-Fold Splitting
• Strict zero patient identity leakage across partitions
• Per-lead z-score normalization: $x_c' = (x_c - \mu_c)/\sigma_c$
• Train (Folds 1–8): $n = 17,411$ (15,082 patients)
• Validation (Fold 9): $n = 2,183$ (1,894 patients)
• Held-Out Test (Fold 10): $n = 2,205$ (1,893 patients)
PHASE 3: Fixed-Budget Architectural Allocation
Arm A: Homogeneous Capacity Scaling ($B \approx 2.0$M)
• Divided equally into K identical ResNet-1D sub-networks (2.0M/K)
• $K = 1$ ($W_1 = 65$, 2.02M) • $K = 2$ ($W_2 = 46$, 2.03M)
• $K = 4$ ($W_4 = 32$, 1.98M) • $K = 8$ ($W_8 = 23$, 2.05M)
• $K = 16$ ($W_{16} = 16$, 2.01M) → High CKA redundancy (0.9458)
Arm B: Parameter-Matched Heterogeneous Arm (1.99M)
• 4 Diverse inductive biases matched to homogeneous $K = 4$ (+0.92%):
⋆ ResNet-1D (494k): 3-stage residual blocks, skip connections
⋆ MultiScale-CNN (509k): Parallel receptive fields (kernels 3, 7, 15, 31)
⋆ TCN-1D (491k): Dilated causal convolutions ($d \in \{1,2,4,8\}$)
⋆ ConvAttention-1D (500k): Downsampling stem + 2-layer Transformer
• Orthogonal feature representations: Mean Linear CKA = 0.9008
PHASES 4 & 5: Adaptive Gating, Stress & Deferral
Phase 4: Dynamic Gating & Sparse Inference
• Patient-Specific Router $g_\phi(x)$ (∼ 16K params, Conv1D + MLP)
• Loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diag}} + 0.05\,\mathcal{L}_{\text{div}} + 0.10\,\mathcal{L}_{\text{fail}}$ (prevents collapse)
• Dense Learned: 414.9 MFLOPs (AUROC 0.9118 ± 0.0004)
• Sparse Top-2 (MOSAIC-Med): 297.8 MFLOPs (−28.2% FLOPs)
Phase 5: Clinical Stress Testing & Selective Deferral
• 6-Stress Benchmark: Clean baseline (0.9113), EMG noise, wander, precordial loss (V1–V3), limb loss (I–III), external site shift ($s \neq 0$)
• Selective Deferral: Uncertainty $U(x) = \mathcal{H}(\hat{y}) + 2.0\,\text{Var}_k(\hat{y}_k)$
Deferring 10% most uncertain ($c = 90\%$): risk drops 11.95% → 10.37%
Empirical relative risk reduction: 4.18% ($p = 0.0093$, 5 seeds)